Can Large Language Models Pass IFoA Professional Examinations?

Abstract

Gemini 3.1 Pro in high-reasoning mode sat 28 IFoA professional examinations from the September 2025 sitting and passed all of them, achieving a 100% pass rate under three grading conditions (Strict, Fair, and Generous). The average score under Fair grading was 87.6%, with an average margin of +30.1 percentage points above the published pass marks.

Both the exam-taking and grading phases were performed by the same model, introducing an inherent self-assessment bias that is partially mitigated by grading each paper three ways and by human validation of 27 question parts across four papers. Under the most punitive Strict grading, five exams passed with a margin of 5 percentage points or less. These are single-run results from one model on one sitting; the findings should be treated as indicative rather than definitive.

Exams are only one layer of the IFoA's qualification framework, which also requires supervised work experience, ethical obligations, peer review, Practising Certificates, and Continuing Professional Development. This paper aims to prompt discussion within the profession on the role AI might play in actuarial work.

1. Introduction

The IFoA administers a rigorous series of examinations to ensure that practitioners possess the mathematical, statistical, and commercial knowledge required to manage financial risk. These exams have historically stood as a high barrier to entry, testing not only technical knowledge but also the application of complex principles to novel scenarios.

The rapid advancement of Large Language Models (LLMs) has prompted a re-evaluation of cognitive and analytical work across many professions. Recent reasoning-focused models have demonstrated significant improvements in logical reasoning, multi-step derivations, and professional-quality written analysis.

The question here is whether a modern LLM can meet the technical standard set by the IFoA across the full breadth of its qualification syllabus. As a pensions actuary and IFoA Fellow, my interest is practical: understanding where AI sits relative to the professional standard, what it gets right and wrong, and what the results mean for the profession. The study covers 28 exam papers spanning all four qualification stages, providing clear, empirical data on current AI capabilities within an actuarial context.

2. Methodology

2.1 Scope and Exam Selection

The experiment covered 28 complete exam papers from the September 2025 sitting of the IFoA examinations. This sitting was specifically chosen because the papers were published after the model's training data cutoff (January 2025), eliminating the risk of data contamination.

Stage	Subjects	Papers
Core Principles	CB1, CB2, CM1 (A+B), CM2 (A+B), CS1 (A+B), CS2 (A+B)	10
Core Practices	CP1 (Papers 1+2), CP2 (Papers 1+2), CP3	5
Specialist Principles	SP1, SP2, SP4, SP5, SP6, SP7, SP8, SP9	8
Specialist Advanced	SA1, SA2, SA3, SA4, SA7	5

For subjects with more than one paper, each was attempted and graded independently.

2.2 Model Selection and Configuration

The model used for both the exam-taking and grading phases was Gemini 3.1 Pro in its high-reasoning configuration ("High" mode), which enables extended chain-of-thought processing before producing a response.

All experiments were run using Google Antigravity's standard chat mode with the "High" reasoning setting selected. No manual parameter overrides (temperature, token limits, etc.) were applied. The exact values of the platform defaults were not recorded and are not exposed by the interface, meaning the experiment cannot be exactly reproduced, only approximately replicated. This limitation is discussed further in Section 5.

2.3 Experimental Architecture

The experiment was orchestrated within Google Antigravity, a publicly available agentic coding and automation environment developed by Google. A dual-agent architecture was used, simulating the candidate-examiner dynamic:

1. The Exam-Taker Agent. Provided with the exam question paper and instructed to act as an actuarial candidate. During the attempt phase, the agent had access only to the exam question paper (converted from PDF to Markdown), the official IFoA Formulae and Tables book (also converted to Markdown), any provided Excel workbooks or R data files, its own pre-trained knowledge. The examiner's report, marking scheme, and solutions material were excluded.

2. The Exam-Grader Agent. A separate instantiation of the same model, subsequently provided with three inputs: the question paper, the LLM's attempt, and the official IFoA examiner's report. It evaluated the attempt against the published marking scheme, awarding marks question-by-question.

Both agents used the same underlying model (Gemini 3.1 Pro High). While they operated as independent sessions with no shared context, the use of the same model family for self-assessment introduces a potential bias, discussed in Section 5. Each exam was attempted in a fresh session with no cross-contamination between papers.

2.4 Grading Framework: Tri-Persona Assessment

To account for the inherent subjectivity in marking and to provide a confidence interval around scores, the exam-grader evaluated each paper under three concurrent marking personas:

- **Strict (Harsh):** Marks awarded only when the answer directly matches the specific keywords, phrasing, or methodology in the examiner's report. No benefit of the doubt. This represents a lower bound.
- **Fair (Standard):** Marks awarded for answers that are mathematically equivalent, logically sound, or demonstrate clear understanding, even if the wording differs from the model answer.
- **Generous (Lenient):** Marks awarded for any valid, insightful point, including those beyond the official marking scheme. This represents an upper bound.

Each question part was graded sequentially across all three personas for consistent comparison. The strict persona was designed as a stress test: a deliberately conservative lower bound that, if passed, would confirm the model's knowledge regardless of whether every answer matched the examiner's exact phrasing. The human validation in Section 2.7 tested whether that lower bound held.

2.5 Data Preparation: PDF to Markdown Conversion

IFoA exam papers and examiner reports are published as PDFs. These were converted to Markdown using pypdf to enable clean text processing by the LLM. This conversion is imperfect: mathematical notation, complex tables, and formatted formulae can be garbled during extraction, particularly for CM and CS papers. Importantly, this limitation works against the model (it passed despite receiving garbled inputs on some mathematical papers), suggesting the reported scores may be conservative.

2.6 Exam Execution Environment

Practical and computer-based exams were not merely answered theoretically. The exam-taker agent had access to a live execution environment:

- **Excel-based exams** (CM1B, CM2B, CP2): The agent wrote Python scripts (openpyxl) implementing year-by-year actuarial calculations (unit fund projections, profit vectors, zeroisation) and executed them to produce exact numerical results.
- **R-based exams** (CS1B, CS2B): The agent wrote R scripts for statistical analyses (GLMs, time series, survival analysis, simulation). R execution was less reliable than Python: some scripts ran successfully, but others failed silently due to environment

limitations. Questions requiring interpretation of specific R output received only partial credit where the code was correct but the output was not captured.

- **Modelling exams** (CP2): The agent built complete Excel financial models programmatically, including audit trails, parameter sheets, and automated checks.

All Python scripts were executed and their outputs captured. R execution was partial, with the results for CS papers reflect a mix of executed outputs and theoretically correct but unexecuted code.

2.7 Human Validation

The human validation tested whether the Strict marker functions as a reliable lower bound, specifically whether it might in some cases award *more* marks than a human reviewer would, undermining its value as a stress test.

I independently marked 27 question parts from four papers, selected within my subject expertise as a pensions actuary: CB1 (borderline, +1%), SP4 (+5%), SA4 (+4%), and SP5 (+25%, as a calibration check). The review comprised 19 "disputed" questions where the Strict and Fair markers disagreed materially, and 8 "calibration" checks where both agreed on a high score.

Key findings: - Of 19 disputed questions, Strict was correct or closer on 7 (37%), Fair on 6 (32%), and 5 fell in between. In one case (SA4 Q2(iii)), both markers over-awarded: Strict gave 8/10, Fair gave 10/10, the reviewer justified only 6-6.5/10. - Of 8 calibration checks, 7 confirmed the AI markers were right to award high marks. - After adjustments, all four papers still passed. CB1's lower bound rose from 61% to 67%, confirming that Strict was systematically too harsh on that paper's essay-style questions.

Across 26 of 27 question parts reviewed, the Strict marker did not over-award. The one exception (SA4 Q2(iii)), where plausible confidence masked shallow coverage, is the kind of error self-grading is least equipped to catch. Per-question verdicts are available in the project repository.

3. Results

3.1 Overall Performance Summary

The model passed all 28 exams under all three grading conditions.

Table 1: Overall performance summary across all 28 IFoA exam papers.

Metric	Strict	Fair	Generous
Exams Passed	28/28	28/28	28/28
Pass Rate	100.0%	100.0%	100.0%
Average Score	77.1%	87.6%	93.6%
Average Margin vs Pass Mark	+19.6%	+30.1%	+36.1%

3.2 Detailed Results by Exam

Pass marks are set by the IFoA per sitting and vary by subject (45%-64% in this sitting). Figure 1 provides a visual overview.

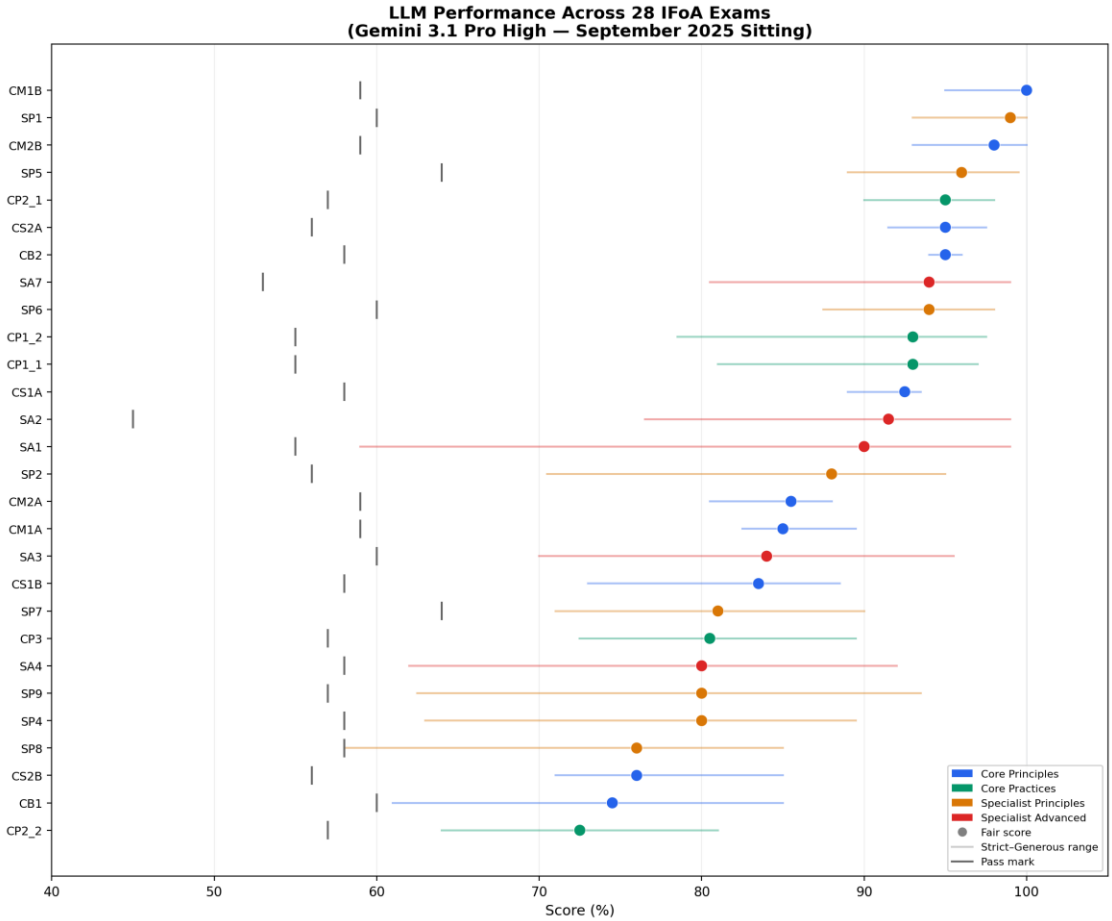

Figure 1: LLM performance across all 28 IFoA exams, showing Fair score (dot), Strict-Generous range (line), and pass mark (grey bar) for each paper.

Table 2: Individual exam results by subject.

Subject	Pass Mark	Strict	Fair	Generous	Strict Margin
Core Principles					
CB1	60%	61.0%	74.5%	85.0%	+1.0%
CB2	58%	94.0%	95.0%	96.0%	+36.0%
CM1A	59%	82.5%	85.0%	89.5%	+23.5%
CM1B	59%	95.0%	100.0%	100.0%	+36.0%
CM2A	59%	80.5%	85.5%	88.0%	+21.5%
CM2B	59%	93.0%	98.0%	100.0%	+34.0%
CS1A	58%	89.0%	92.5%	93.5%	+31.0%
CS1B	58%	73.0%	83.5%	88.5%	+15.0%
CS2A	56%	91.5%	95.0%	97.5%	+35.5%
CS2B	56%	71.0%	76.0%	85.0%	+15.0%
Core Practices					
CP1_1	55%	81.0%	93.0%	97.0%	+26.0%
CP1_2	55%	78.5%	93.0%	97.5%	+23.5%
CP2_1	57%	90.0%	95.0%	98.0%	+33.0%
CP2_2	57%	64.0%	72.5%	81.0%	+7.0%
CP3	57%	72.5%	80.5%	89.5%	+15.5%
Specialist Principles					
SP1	60%	93.0%	99.0%	100.0%	+33.0%
SP2	56%	70.5%	88.0%	95.0%	+14.5%
SP4	58%	63.0%	80.0%	89.5%	+5.0%
SP5	64%	89.0%	96.0%	99.5%	+25.0%
SP6	60%	87.5%	94.0%	98.0%	+27.5%
SP7	64%	71.0%	81.0%	90.0%	+7.0%
SP8	58%	58.0%	76.0%	85.0%	+0.0%
SP9	57%	62.5%	80.0%	93.5%	+5.5%
Specialist Advanced					
SA1	55%	59.0%	90.0%	99.0%	+4.0%
SA2	45%	76.5%	91.5%	99.0%	+31.5%
SA3	60%	70.0%	84.0%	95.5%	+10.0%
SA4	58%	62.0%	80.0%	92.0%	+4.0%
SA7	53%	80.5%	94.0%	99.0%	+27.5%

"Strict Margin" = Strict score minus pass mark.

3.3 Robustness Analysis

The margins under Strict grading reveal varying degrees of confidence:

- **Comfortable passes (Strict margin > 15%):** 16 of 28 exams (57%). Robust even with significant downward adjustment.
- **Moderate passes (Strict margin 5-15%):** 7 exams (25%) - CP2_2, CP3, CS1B, CS2B, SA3, SP2, SP7.
- **Borderline passes (Strict margin < 5%):** 5 exams (18%) - SP8 (+0.0%), CB1 (+1.0%), SA1 (+4.0%), SA4 (+4.0%), SP4 (+5.0%).

SP8 is the most precarious result: the Strict score exactly equals the 58% pass mark. A single mark deducted would change this to a fail.

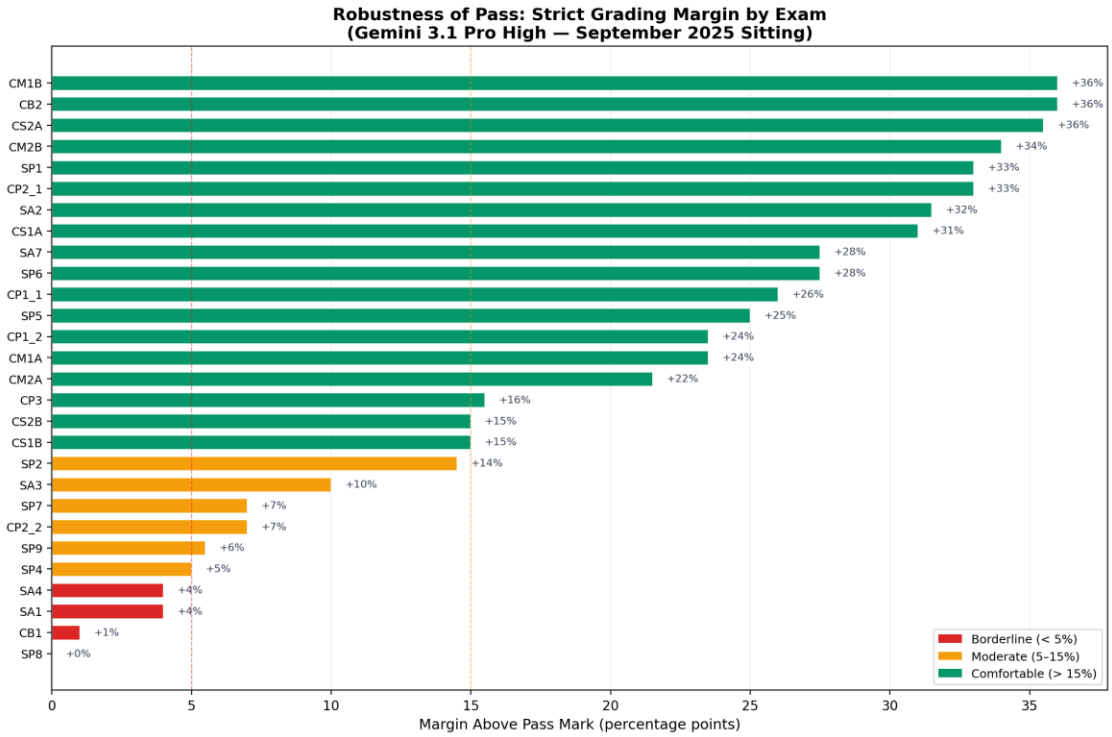

Figure 2: Strict grading margin by exam, colour-coded by robustness. Red bars indicate borderline passes where the result is sensitive to grading assumptions.

3.4 The Role of the Strict Marker as a Stress Test

The human validation confirmed that the Strict marker broadly holds as a lower bound, but tends to err by penalising valid work for the wrong reasons rather than by being too lenient. For example, in CM2A Question 6(ii), a PDF-to-Markdown extraction artefact caused the LLM to work with a different but mathematically equivalent parameterisation. The derivation and final answer were identical. The Strict marker awarded 0/2 for “the wrong parameterisation”; the Fair marker awarded 2/2.

Where the two scores converge (predominantly on mathematical papers) the grading is straightforward. Where they diverge, the gap sometimes reflects literal keyword-matching rather than a genuine deficiency in the answer.

3.5 Performance by Subject Area

- **CB (Business):** 61.0%-94.0% Strict. CB2 was a standout; CB1 was borderline under Strict grading, penalised for not using specific mark-scheme keywords.
- **CM (Mathematics):** Consistently strong (80.5%-95.0% Strict). CM1B achieved 100% Fair. The model navigated multi-state processes, yield curves, and joint-life assurance calculations with precision.
- **CS (Statistics):** Strong on Paper A (89.0%-91.5% Strict), lower on Paper B (71.0%-73.0% Strict) where code execution was required.
- **CP (Practice):** Variable. CP2_1 (modelling) scored 90.0% Strict; CP2_2 (communication) was the weakest Fair result at 72.5%.
- **SP (Specialist Principles):** Wide range from SP8 (58.0% Strict, borderline) to SP1 (93.0% Strict). Performance correlated with the balance of calculation vs. discursive content.
- **SA (Specialist Advanced):** The model handled advanced technical content well but fell short on exam technique, with answers sometimes pitched at the wrong level or too brief for SA-level mark allocations.

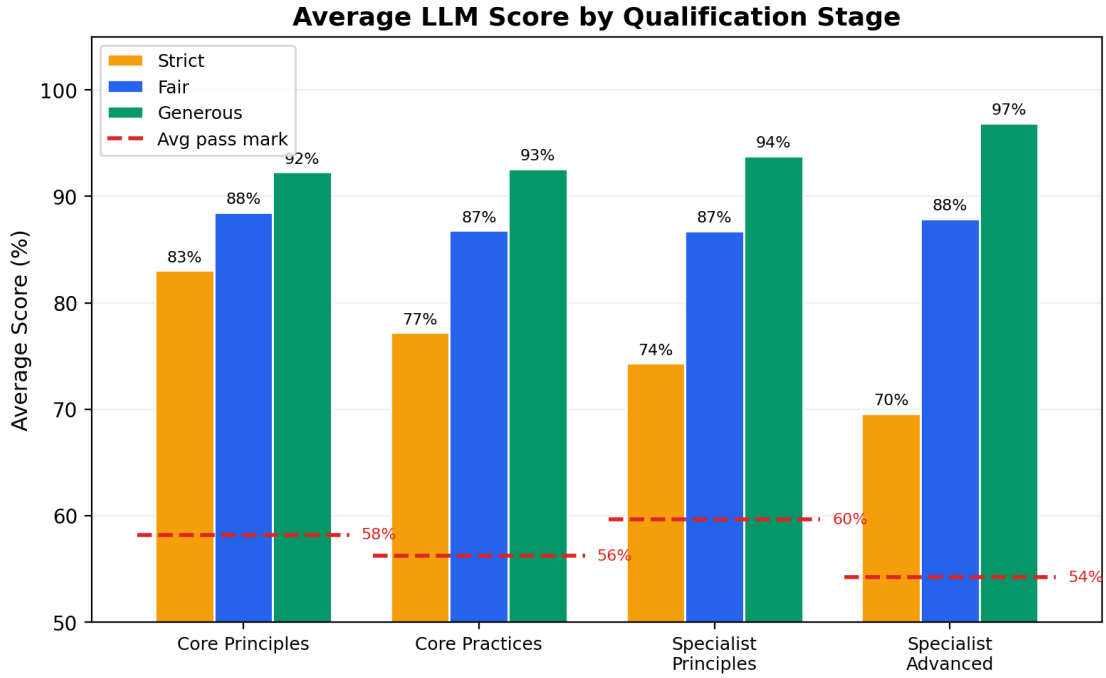

Figure 3: Average LLM score by qualification stage. The Strict score drops at higher stages (reflecting more subjective marking), but the Fair score remains consistent at 87-88% across all four stages.

3.6 Performance by Question Type

Table 3: Summary of model performance by question type.

Question Type	Typical Performance	Notes
Multiple choice	Moderate to High	Strong knowledge base; occasionally over-analyses simple questions
Short calculation	Very High	Precise formula application and accurate results
Long calculation	High	Sound methodology; occasional transcription or magnitude errors
Bookwork / definitions	High	Accurate but sometimes generic rather than exam-specific
Discussion / essay	High (with variance)	Strong commercial insight; can miss specific examiner bullet points
Mathematical proofs	Very High	Rigorous derivations with clear logical steps
R/Excel practical	High	Actual code execution; results are computed, not estimated

4. Analysis and Observations

4.1 Mathematical and Technical Reasoning

The model's strongest area was mathematical reasoning. It consistently produced correct derivations for complex actuarial calculations, including multi-state transition models, forward rate structures, joint-life assurance functions, and unit-linked contract valuations.

A striking finding is how consistently the model avoided the specific errors that examiner reports identify as common traps for human candidates. In CM1B, it correctly applied backward recursive zeroisation (a procedure the examiner noted most candidates failed to use properly). In CM2B, it computed E[U(V)] as the mean of the utility of each path rather than the utility of the mean, a distinction the examiner flagged as catching many candidates. These represent the kind of technically precise reasoning that separates understanding from memorisation.

Where errors occurred they were almost always traceable to PDF parsing artefacts introducing the wrong formula or parameter value (CM1A, CM2A). The model showed no systematic weakness in any particular type of mathematical reasoning.

4.2 Essay Questions

The model's performance on essay questions was markedly more variable. Four distinct failure modes emerge from the grade reports.

Brevity and level mismatch. The model consistently answered at the wrong level of abstraction. On SA4 Q1(v), a 12-mark question, the model produced a technically valid exposition of IFA capacity, tax complexities, and regulatory guidance, but the examiner wanted higher-level discussion of scheme funding impacts. In SP4 Q2, the Strict marker noted a "scattergun problem": the model covered every plausible consideration but spread its marks too thinly to accumulate the specific high-value points. This pattern appeared in most SA and SP papers.

Missing examiner-specific phrases. The model demonstrated understanding without stating the specific phrase the examiner required. In SA4 Q2(ii), it gave a correct explanation of the actuarial control cycle, prudence, and successive correction, but never stated the core point: that assumptions do not affect the actual benefits paid.

Skipping the setup. The model tended to jump straight to conclusions without first defining terms or explaining concepts. IFoA marking schemes award marks for explanation as well as the conclusion. In SA2 Q3(iv), it provided the corrective action without first identifying the fault, losing marks despite the substance being right.

Textbook answer versus applied context. Less common but producing large mark drops: the model applying the right framework to the wrong context. In SA3 Q2(v), it answered a retroactive reinsurance question with forward-looking reinsurance management, scoring 0/3.

4.3 Practical and Computer-Based Assessment

The model performed strongly on structured modelling papers. CM1B (95% Strict, 100% Fair) and CM2B (93% Strict, 98% Fair) were handled end-to-end using Python scripts, with correct actuarial sequences throughout. In several instances the model replaced iterative Excel methods such as Goal Seek with direct algebraic solutions, a valid full-mark approach under Fair grading.

Two limitations emerged. First, a tooling constraint: while the model wrote correct R code throughout, the execution environment did not reliably support R, so CS paper questions requiring interpretation of specific output received only partial credit. Second, a data interpretation gap: CP2_2 (64% Strict, the weakest practical result) missed a time series seasonality component and plotted absolute prices where the question asked for price changes. The model is more reliable with clearly defined computational tasks than with exploratory analysis.

4.4 A Characteristic Pattern

The model reliably avoids the technical errors that catch human candidates, while reliably missing the specific phrases and framing that examiners expect in essay answers. A human candidate with strong exam technique knows to pitch answers at the right level and structure them around mark scheme headings, but may lack technical precision. The model's profile is the inverse: strong technical precision, limited intuition about what an examiner wants from a six-mark discussion question. On mathematical papers, this profile

is near-optimal. On discursive questions, the failure mode is often not a lack of knowledge but a mismatch in framing.

5. Limitations and Threats to Validity

5.1 Internal Validity: Self-Grading Bias

The same model family was used for both exam-taking and grading. LLMs have been shown to exhibit self-preference in evaluation tasks, with research on LLM-as-judge frameworks finding that self-evaluation scores can be inflated by 5-15% relative to human evaluation[1], with the bias most pronounced on open-ended and subjective tasks[2].

The human validation's most telling finding was SA4 Q2(iii), where both AI markers significantly over-awarded relative to the human reviewer, with plausible confidence masking shallow coverage. This is consistent with the expected direction of self-preference bias and most likely to appear on discursive questions, which is precisely where the Strict-Fair gap is widest. Across the 26 other question parts reviewed, the Strict marker did not over-award. Independent human re-grading of the full set would provide a more reliable assessment.

5.2 External Validity: Generalisability

One model, one exam sitting, one run per exam. No claims of statistical significance can be made from n=1. Results may not generalise to other sittings, other models, or repeat runs.

5.3 Construct Validity

Exam scores are a proxy for competence, not competence itself. The IFoA's own framework treats exams as necessary but not sufficient (see Section 6.1). The model's scores demonstrate technical knowledge, not professional readiness.

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[1] Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E.P., Zhang, H., Gonzalez, J.E., and Stoica, I. (2023). "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena." *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, Datasets and Benchmarks Track. Available at: https://arxiv.org/abs/2306.05685

[2] Panickssery, A., Bowman, S.R., and Feng, S. (2024). "LLM Evaluators Recognize and Favor Their Own Generations." *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*. Available at: https://arxiv.org/abs/2404.13076

5.4 PDF-to-Markdown Conversion Artefacts

Mathematical notation and formatted formulae can be garbled during PDF extraction, affecting both the model's inputs and the grader's marking scheme. This limitation works against the model, meaning reported scores are likely conservative.

5.5 Reproducibility

The experiment was not re-run. Unrecorded model defaults and a specific platform version mean exact replication is not possible. Future work should include multiple runs to establish score distributions and confidence intervals.

6. Discussion: Implications for the Profession

The results demonstrate that a modern LLM can meet the technical standard of the IFoA examination syllabus. But passing exams is a long way from being a qualified actuary.

6.1 Exams Are One Layer of a Much Deeper System

Exams are the most visible layer of the IFoA's quality framework, but they sit within a system of interlocking requirements:

Requirement	Summary
Supervised work experience (PPD)	24+ months supervised experience; 30+ competencies signed off by a workplace supervisor
Continuing Professional Development	15 hours/year minimum (including 2 on ethics); subject to audit
Practising Certificates	Required for reserved roles (Scheme Actuary, Chief Actuary, etc.); personal regulatory accountability
Actuaries' Code	Six principles including integrity, speaking up, and impartiality; require moral agency
Peer review (APS X2)	Mandatory, proportionate to difficulty and consequences
Quality Assurance Scheme (QAS)	Organisational accreditation; six-year cycle
External oversight	FRC, PRA, FCA oversight; IFoA Disciplinary Scheme

The profession has never treated passing exams as proof that someone is ready to sign off work alone. Every one of these layers exists because exam-level competence is necessary but not sufficient.

6.2 Passing Is Not the Same as Being Right

Under strict marking, five exams passed with a margin of less than 5 percentage points, meaning the model was arguably wrong or incomplete on 35-40% of the content. But the pass marks ranged from 45% to 64%; the profession does not consider a newly qualified actuary who just cleared the pass mark ready to work unsupervised either. An LLM that passes exams sits in a similar position to a newly qualified actuary: it can produce useful technical work, but is not trustworthy as the sole author of advice carrying professional responsibility.

6.3 AI as a Team Member, Not a Replacement

The more practical question is whether AI can contribute as a team member within existing professional frameworks.

The model's strong technical performance suggests it could produce initial drafts of actuarial reports, model documentation, or advice letters, with that work that a qualified actuary would review and refine under APS X2, much as junior team members contribute today. The grading phase demonstrated its ability to evaluate work against a detailed marking scheme, a capability that could support peer review, catching mathematical errors and flagging inconsistencies a time-pressed reviewer might miss. The practical exams showed it can build complete models, write and execute computation scripts, and compile research for assumption-setting. As LLMs become integrated with enterprise environments, including running Python, R, and Excel against live datasets, the distinction between theoretical and practical AI capability narrows.

None of these applications replace the qualified human who carries professional responsibility. All work within the existing framework of review, sign-off, and accountability.

6.4 Accountability and the Broader Professional Requirements

The deeper question is one of accountability. Many PPD competencies, including the application of technical knowledge, clear communication of results, and analytical problem-solving, appear feasible given the capabilities demonstrated here. Others, such as navigating the politics of a difficult board meeting, remain beyond current AI. But the Practising Certificate and Disciplinary Scheme presuppose personal accountability by an identifiable individual, which is a legal and regulatory requirement rather than a technical one. The profession's quality framework is built on the principle that identifiable people must be accountable for work that affects the public. Whether that framework evolves to accommodate AI tools is a question for the profession, its regulators, and the public it serves.

6.5 Lessons from Other Professions

The actuarial exams are not the first professional examinations passed by AI. GPT-4 passed the Uniform Bar Exam at approximately the 90th percentile[3] and the USMLE Steps 1-3 with accuracy rates of 92-95%[4]. The responses from those professions reveal common threads: mandatory disclosure of AI use in professional work (Pennsylvania courts, California's AB 3030)[5]; AI literacy added to continuing education requirements; exam redesign toward skills-based assessment (the NextGen Bar Exam); and integration rather than prohibition.

The IFoA's current exams heavily weight technical derivation and knowledge recall, capabilities where LLMs perform strongly. If an AI can reliably derive a joint-life assurance function or explain the dynamics of a CIR model, the exam's value shifts from testing whether a candidate *can* do these things to testing whether they know *when* to do them, *why*, and what the limitations are. Professional judgement, ethical reasoning, and the critical evaluation of model outputs may deserve greater weight in future syllabi.

6.6 The Profession Is Already Engaging with AI

The IFoA's AI Ethics Working Party has published practical guidance on managing LLM hallucination[6], an ethical checklist for AI models[7], and a framework for controlling LLM use in practice[8]. The working party has argued that actuaries are uniquely positioned to lead AI

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[3] Katz, D.M., Bommarito, M.J., Gao, S., and Arredondo, P. (2024). "GPT-4 passes the bar exam." *Philosophical Transactions of the Royal Society A*, 382(2270), 20230254. DOI: 10.1098/rsta.2023.0254

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[5] Pennsylvania Bar Association and Philadelphia Bar Association (2024). "Joint Formal Opinion 2024-200: Ethical Issues Regarding the Use of Artificial Intelligence." 22 May 2024.

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[8] Huang, F. (2025). "Check your AI: a framework for its use in actuarial practice." *The Actuary*, June 2025. Available at: https://www.theactuary.com/features/2025/06/25/check-your-ai-framework-its-use-actuarial-practice

governance, given their expertise in modelling, uncertainty, and risk[9]. An AI that passes exams but occasionally hallucinates is precisely the kind of tool that requires these governance frameworks. The profession's proactive engagement positions it well to integrate AI tools thoughtfully rather than reactively.

7. Conclusion

A modern LLM can pass the IFoA actuarial examinations across all four qualification stages. The 100% pass rate across 28 papers, with an average Fair score of 87.6%, demonstrates substantial capability in mathematical reasoning, statistical analysis, and commercial judgement , subject to material caveats around self-grading, borderline margins, and the inherent limitations of a single-sitting, single-run experiment.

Passing exams is not the same as being a competent actuary. The IFoA's qualification framework layers supervised experience, ethical obligations, peer review, and regulatory oversight on top of examinations: none of which an AI system can currently satisfy.

What I think the results actually show is that technical knowledge is no longer the hard part for AI. If AI can reliably demonstrate the mathematical and technical content, it may be worth considering whether the skills that are harder to automate, such as professional judgement, challenging a client, and knowing when a model output feels wrong, deserve greater weight in the assessment framework going forwards. Actuaries, with their deep expertise in modelling, uncertainty, and risk, are well placed to lead that conversation.

Future work:

- Independent human re-grading of the full paper set (particularly the five borderline papers) would confirm whether the 100% pass rate holds.
- Refining the exam-taker prompting to address the discursive failure modes identified in Section 4.2 would test how much of the Strict/Fair gap is due to exam technique.
- Replication across additional sittings and models, including the use of a different model family for grading, would address the generalisability and self-grading bias limitations.

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[9] Bhattacharyya, A. (2025). "Why actuaries are uniquely positioned to lead AI governance." **Institute and Faculty of Actuaries Blog**, 22 August 2025. Available at: https://blog.actuaries.org.uk/2025/8/why-actuaries-are-uniquely-positioned-to-lead-ai-governance/

Disclosure: This is a personal research project. The author is a Fellow of the IFoA and a pensions actuary. No external funding was received. Full methodology, grade reports, and raw data are available in the project repository. Exam content remains the intellectual property of the IFoA.

Experiment conducted: February-March 2026.

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